section will provide a solution to accomplish that.

13.1.2 Managing logs with Loki, Fluent Bit, and Grafana

When you move to distributed systems like microservices and complex environments like the cloud, managing logs becomes challenging and requires a different solution than in more traditional applications. If something goes wrong, where can we find data about the failure? Traditional applications would rely on log files stored on the host machine. Cloud native applications are deployed in dynamic environments, are replicated, and have different life spans. We need to collect the logs from all applications running in the environment and send them to a central component where they can be aggregated, stored, and searched.

There are plenty of options for managing logs in the cloud. Cloud providers have their own offerings, like Azure Monitor Logs and Google Cloud Logging. There are also many enterprise solutions available on the market, such as Honeycomb, Humio, New Relic, Datadog, and Elastic.

For Polar Bookshop, we'll use a solution based on the Grafana observability stack (https://grafana.com). It's composed of open source technologies, and you can run it yourself in any environment. It's also available as a managed service (Grafana Cloud) offered by Grafana Labs.

The components of the Grafana stack we'll use for managing logs are Loki for log storage and search, Fluent Bit for log collection and aggregation, and Grafana for log data visualization and querying.

NOTE Which technology you use for managing logs is a platform choice and shouldn't impact the applications at all. For example, you should be able to replace the Grafana stack with Humio without making any changes to the Polar Bookshop applications.